

Simple Square-Wave Generator

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The standard bench-top signal generator used in a lab comes with limited output channels and is expensive. So it is highly desir-

able to have a simple low-cost square-wave generator that can be built quickly with easily-available components. Presented here is a simple circuit built around two NE555 timer ICs to generate a variable square-wave signal.

Circuit and working

The circuit diagram of a simple low-cost square-wave generator with bipolar and unipolar outputs is shown in Fig. 1. The circuit is built around step-down transformer X1, bridge

TABLE I
Position of Switch S2 and Corresponding Frequency at TP3

Range	S2 Position
6kHz-60kHz	1
600Hz-6kHz	4
60Hz-600kHz	3
6Hz-60Hz	2

TABLE II
Position of Switch S3 and Corresponding Period at TP4

Duration	S3 Position
11 μ s-170 μ s	1
110 μ s-1.7ms	4
1.1ms-17ms	3
11ms-170ms	2

TABLE III
Test Points

Test point	Details
TP0	0V, GND
TP1	+6V
TP2	-6V

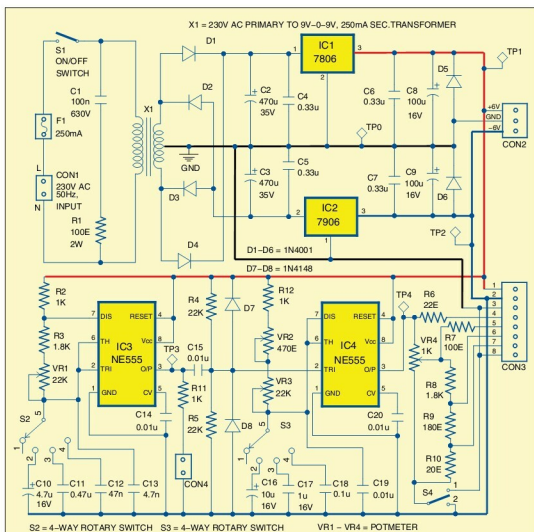


Fig. 1: Circuit diagram of the simple square-wave generator

rectifier comprising diodes D1 through D4, regulators 7806 (IC1) and 7906 (IC2) and two NE555 timer ICs (IC3 and IC4). The timer ICs can work with bipolar power supply as shown in Fig. 1. In this case, the power supply will be $\pm 6V$, produced by linear regulators IC1 and IC2.

Here IC3 is a square-wave generator with four frequency ranges controlled with the position of switch S2. The approximate frequency ranges available are shown in Table I. The frequency within a particular range is varied with VR1.

The pulses produced by timer IC3 are differentiated by C15, R4 and R5, which start the timer IC4. Here, IC4 works as a monostable. It produces a single pulse